

CONTACT INFORMATION	Daniel J. Epstein Dept. of Industrial and Systems Engineering, Viterbi School of Engineering 3715 McClintock Avenue Los Angeles, CA 90089-0193	Website febattista.github.io Email f.battista95@gmail.com
RESEARCH INTERESTS	Discrete optimization, spanning theory, algorithms, and computation in the development of exact solution methods, with emphasis on branch and cut for complex mixed-integer linear problems, including those arising in bilevel optimization and combinatorial optimization, and on the development of open-source tools for the community.	
ACADEMIC EXPERIENCE	University of Southern California, Daniel J. Epstein Dept. of Industrial and Systems Engineering, Los Angeles, CA, USA Postdoctoral Researcher under Karmel S. Shehadeh February 2026 – Present Lehigh University, Dept. of Industrial and Systems Engineering, Bethlehem, PA, USA Postdoctoral Researcher under Ted Ralphs February 2023 – January 2026	
EDUCATION	Università di Roma Sapienza Dept. of Computer, Control, and Management Engineering, Rome, Italy Ph.D. in Operations Research November 2019 – January 2023 Thesis: On Semidefinite Lift-and-Project Relaxations of Combinatorial Optimization Problems Advisors: Marianna De Santis, Fabrizio Rossi, Stefano Smriglio Alpen-Adria-Universität Institut für Mathematik, Klagenfurt, Austria Visiting Ph.D student under Angelika Wiegele February 2022 – July 2022 Università degli Studi dell'Aquila Dept. of Information Engineering, Computer Science and Mathematics, L'Aquila, Italy Master's degree in Computer Science, <i>Magna Cum Laude</i> July 2019 Bachelor's degree in Computer Science, <i>Magna Cum Laude</i> October 2017	
REFEREED PUBLICATIONS	[1] F. Battista, F. Rossi, and S. Smriglio. Application of the Lovász-Schrijver Operator to Compact Stable Set Integer Programs. In: <i>Mathematical Programming Computation</i> (2025). DOI: 10.1007/s12532-025-00298-8. [2] F. Battista and M. De Santis. Dealing with Inequality Constraints in Large-Scale Semidefinite Relaxations for Graph Coloring and Maximum Clique Problems. In: <i>4OR. A Quarterly Journal of Operations Research</i> (2024), pp. 1–31. DOI: 10.1007/s10288-024-00569-5.	
WORKING PAPERS	[3] F. Battista, F. Rossi, and S. Smriglio. Linear versus Semidefinite Lovász–Schrijver Lifting Operators of Stable Set Integer Programs. Working paper. 2026. [4] K. S. Shehadeh, M. Y. Tsang, and F. Battista. Contextual Residual-based Distributionally Robust Optimization under Predictive Model Uncertainty. Working paper. Dept. of Industrial and Systems Engineering, University of Southern California, 2026. [5] F. Battista and T. K. Ralphs. Improving Directions in Mixed Integer Bilevel Linear Optimization. arXiv preprint. COR@L Laboratory, Lehigh University, 2025. URL: https://doi.org/10.48550/arXiv.2511.03566. <i>Under review at Mathematical Programming Computation</i>.	

- [6] S. Fallah, F. **Battista**, and T. K. Ralphs. *A Branch-and-Bound Algorithm for Multiobjective Mixed Integer Linear Optimization*. Working paper. COROL Laboratory, Lehigh University, 2025.

SOFTWARE

- [7] F. **Battista**, F. Rossi, and S. Smriglio. *SDP_lift_and_project* (2025). URL: https://github.com/febattista/SDP_lift_and_project. *Maintainer*.
- [8] F. **Battista** and M. De Santis. *ADAL-ineq* (2024). URL: <https://github.com/febattista/ADAL-ineq>. *Maintainer*.
- [9] S. DeNegre, T. Ralphs, and S. Tahernejad. *MibS Version 1.2* (2024). URL: <https://github.com/coin-or/MibS>. *Contributor*.
- [10] T. Ralphs, M. Güzelsoy, and A. Mahajan. *SYMPHONY Version 5.7* (2023). URL: <https://github.com/coin-or/SYMPHONY/>. *Contributor*.

STUDENTS

Doctoral Students Mentorship

- Shannon Kelley, Lehigh University, ISE, Advisor: Ted Ralphs 2023 – 2026
- Samira Fallah, Lehigh University, ISE, Advisor: Ted Ralphs 2023 – 2026
- Hesam Shaelaie, Lehigh University, ISE, Advisor: Ted Ralphs 2023 – 2026
- Suresh Bolusani, Lehigh University, ISE, Advisor: Ted Ralphs 2023 – 2026
- Yu Xie, Lehigh University, ISE, Advisor: Ted Ralphs 2023 – 2026

TEACHING EXPERIENCE

Teaching Assistant

- ISE 418 Integer Programming, Ph.D. course, Lehigh University 2024
- Operations Research Laboratory, B.Sc. course, Università Roma Sapienza 2020
- Advanced Calculus, B.Sc. course, Università Roma Sapienza 2020

TALKS

11. F. **Battista** and T. K. Ralphs. *The Role Of Improving Directions in Mixed-Integer Bilevel Linear Optimization*. In: *Bilevel Optimization Society Webinar Series* (2026). **Invited Talk**.
12. F. **Battista**, T. K. Ralphs, and S. Fallah. *The Restricted Value Function of MILPs and Its Construction with SYMPHONY*. In: *18th INFORMS Computing Society (ICS) Conference* (2025).
13. F. **Battista** and T. K. Ralphs. *A Branch-and-Cut Algorithm for Mixed-Integer Bilevel Linear Optimization Based on Improving Directions*. In: *International Symposium on Mathematical Programming* (2024).
14. F. **Battista** and T. K. Ralphs. *Exploiting Dual Functions in Mixed Integer Bilevel Linear Programs*. In: *INFORMS Annual Meeting* (2023).
15. F. **Battista**, F. Rossi, and S. Smriglio. *Application of the Lovász-Schrijver Operator to Representative Formulation for Graph Coloring Problem*. In: *International Symposium on Combinatorial Optimization* (2022).
16. F. **Battista** et al. *On Semidefinite Lift-and-Project Relaxations for Combinatorial Optimization Problems*. In: *University of Pavia* (2022).
17. F. **Battista** and M. De Santis. *Dealing with Inequalities in Large Scale Semidefinite Programs*. In: *International Conference on Optimization and Decision Science* (2021).

SERVICE

Editorial Board (Associate Editor)

- *Operations Research* (Area: Data, Software, and Computation) 2025 – Present